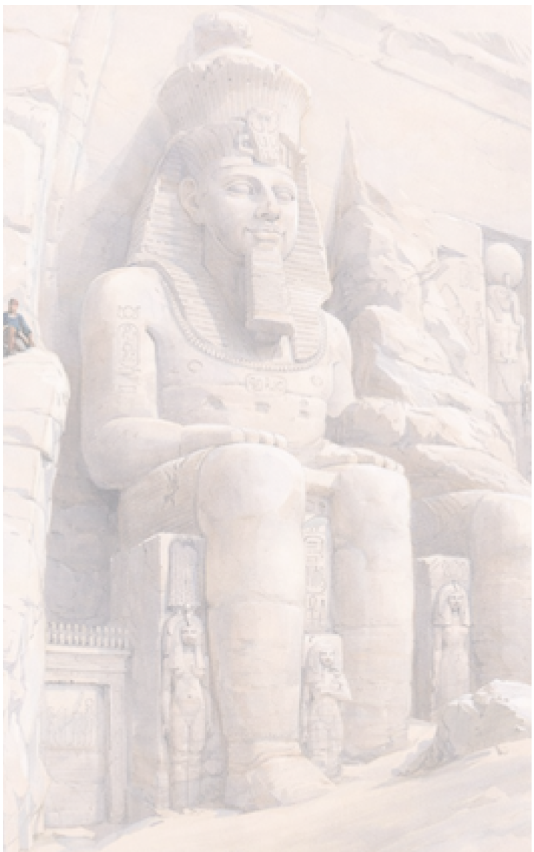
CMPUT 461/561 Fall 2026

Lec 03 -- Parts of Speech

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Parts of Speech

- Parts of Speech and human understanding
- Common Parts of Speech
- Open and closed word classes
- Computer programs to tag words
  - English PoS tagsets
  - Transducers
  - The TBL (Brill) Tagger

Parts of Speech

In grammar, a **part of speech** or **part-of-speech** (abbreviated as **POS** or **PoS**, also known as word class or grammatical category) is a **category of words** (or, more generally, of lexical items) **that have similar grammatical properties**.

Examples:

- A **noun** is a word that represents a concrete or abstract thing, like living creatures, places, actions, qualities, states of existence, and ideas.
- A **verb** is a word that generally conveys an action, an occurrence, or a state of being.

https://en.wikipedia.org/wiki/Part_of_speech

Text interpretation and human understanding

The study of words and their role in language started centuries ago.

- Early philosophers have a profound impact on how we *formulate* and *interpret* tex (containing our ideas and thoughts) in natural language.

In the **6th or 5th century BCE**, the Sanskrit grammarian Yāska defined four main categories of words.

Noun and **verb** were among them, and remain essentially unchanged for 2,500 years!

By the end of the **2nd century BCE**, the Greek grammarian Thrax expanded this classification into eight categories: noun; verb; participle; article; pronoun; preposition; adverb; conjunction;

All are still in use today.

Language and geography

There are over 7,000 living languages in the world (as of 2024). Linguists group them based on common ancestry.

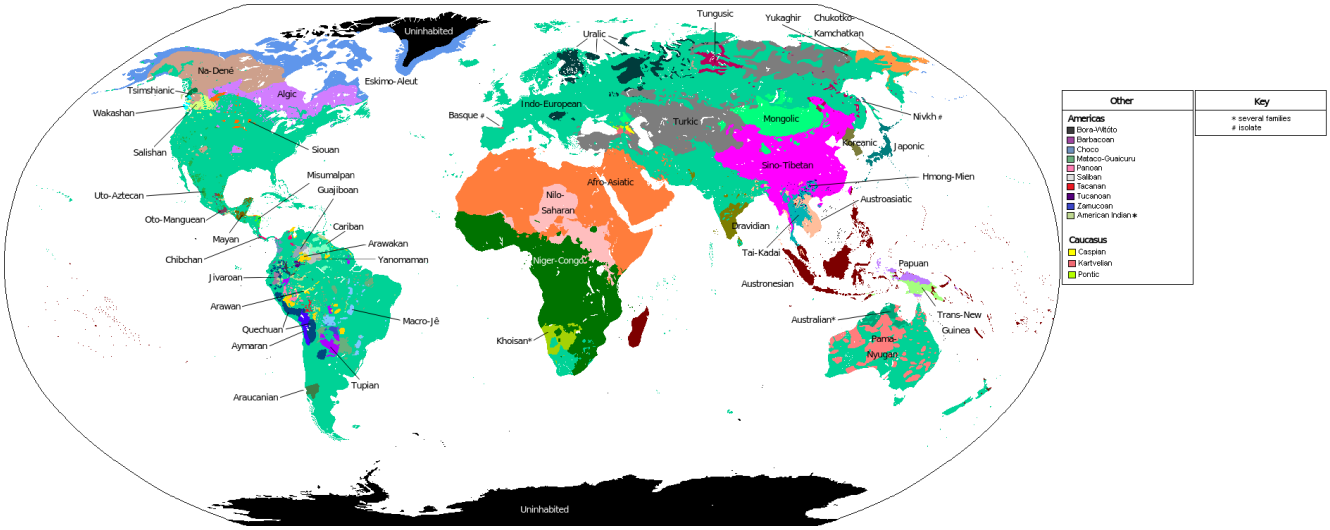
A language **family** is a group of languages having a common ancestor.

All **Romance** languages (ex: Portuguese, French and Italian) descend from *Vulgar Latin*. They belong to the Indo-European group.

Glottolog 5.0

Family	No. of languages
Atlantic–Congo	1,410
Austronesian	1,274
Indo-European	586
Sino-Tibetan	514
Afroasiatic	381
Trans–New Guinea	316
Pama–Nyungan	250
Otomanguean	181
Austroasiatic	158
Tai–Kadai	95
Dravidian	85
Arawakan	77

2005 map of the contemporary distribution of the world's primary language families



https://en.wikipedia.org/wiki/Language_family

Language and thought

We don't know if all (spoken) languages have a single common ancestor.

- Linguists and anthropologists continue to study the origin of the modern human.

Peoples who lived in isolation from other cultures developed languages with unique properties.

Example: Tupi (spoken in modern-day Brazil) has a different concept of time.



LINGUISTICS

Language origin debate rekindled

Eurasian steppe gains ground as Indo-European birthplace.

BY EWEN CALLAWAY

From Icelanders to Sri Lankans, some 3 billion people speak the more than 400 languages and dialects that belong to the Indo-European family. Two fresh studies — one of ancient human DNA, the other a newly constructed genealogical 'tree' of languages — point to the steppes of Ukraine and Russia as the origin of this major language family, rekindling a long-standing debate.

Scholars have long recognized an Indo-European language group that includes Germanic, Slavic and Romance languages as well as classical Sanskrit and other languages of the south Asian subcontinent. Yet the origins of this family of tongues are mired in controversy.

Some researchers hold that an early Indo-European language was spread by Middle Eastern farmers around 8,000–9,500 years ago (see 'Steppe in time'). This 'Anatolian hypothesis' is supported by well-documented migrations into Europe, where agriculturalists replaced or interbred with the existing hunter-gatherers. In 2012, a team led by evolutionary biologist Quentin Atkinson of the University of Auckland in New Zealand produced a family tree of Indo-European tongues that also pointed to an Anatolian origin more than 8,000 years ago.

A competing theory posits that the languages emerged on the Eurasian steppe some 5,000–6,000 years ago, when the domestication of horses and invention of wheeled transport would have allowed herders there to rapidly expand their range. Proponents of the 'steppe hypothesis' note that linguistic reconstructions of a proto-Indo-European tongue include words associated with wheeled vehicles, which were not invented until long after Middle Eastern farmers had reached Europe. "Most linguists have signed up to the steppe hypothesis," says Paul Heggarty, a linguist at the Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany.

One knock against the theory was a lack of compelling evidence for a large-scale migration from the Eurasian steppe at this time.

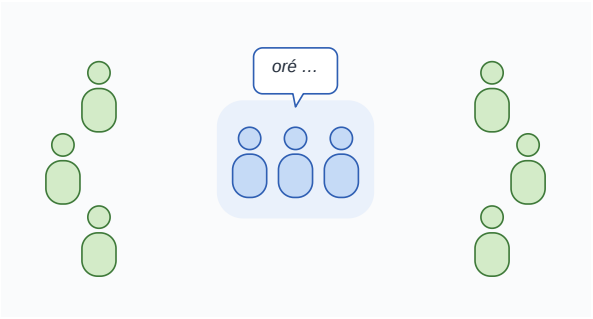
A study of ancient human DNA posted to the bioRxiv.org preprint server on 10 February now plugs that gap (W. Haak *et al.* http://doi.org/z9d; 2015). A team led by David Reich, an evolutionary and population geneticist at Harvard Medical School in Boston, Massachusetts, analysed DNA from the bodies of 94 individuals who lived across Europe between 8,000 and 3,000 years ago. The data confirmed the arrival of Middle Eastern farmers in Europe between 8,000 and 7,000 years ago. But they also revealed evidence for a second migration that began several thousand years later. DNA

Pronouns in Tupi

Tupi was spoken by an indigenous people of modern-day Brazil.

It has only one pronoun in common with English (`ixé` means **I**), and it has two different pronouns for "we":

- `iandé` includes both the speakers and the listeners!
- `oré` excludes the listeners!



Three speakers (in blue) and six listeners.

When saying `oré`, the speaker refers to those having the conversation but not the others who are present and can hear it.

Verbs in Tupi do not express time!

The tense of an action is not expressed by the verb, as in Indo-European languages.

Instead, adjectives and nouns had temporal inflection (`rama`-future; `puêra`-past):
`ybyrá` "tree" `ybyrárama` "a future tree"
`ybyrápuêra` "what was once a tree"

When needed, tense could also be indicated by adverbs like `ko ara`, "this day".

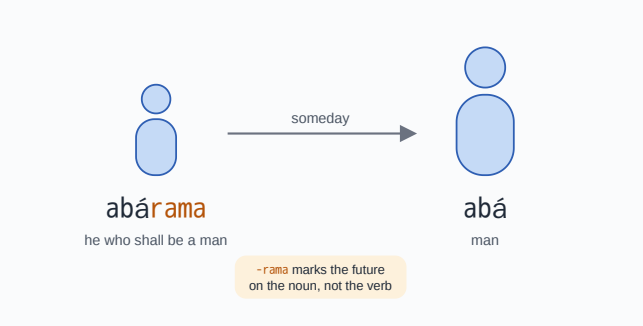
The clitic `-ne` is always added to the last word in the sentence, independent of its grammatical class.

- `Aker pytuna` (I sleep in the forest)
- `Aker pytunane` (I will sleep in the forest)

Do we all think in the same way?

A 14-year-old Tupi-speaking boy **says** that he **is** `abárama` ("he who shall be a man someday").

A 14-year old English-speaking boy **says** that he **will be** a man someday.



Do these two boys think in the same way?

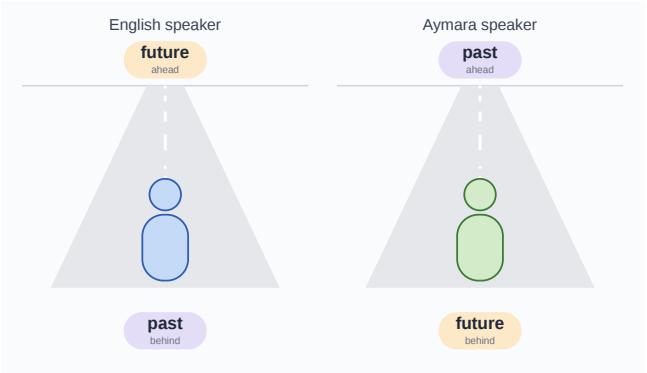
Does the language influence how we think and behave?

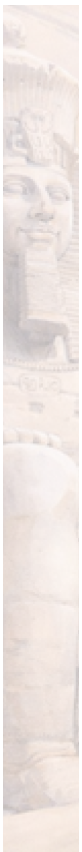
Future and past in Aymara

Language spoken in the Andean highlands of western Bolivia, southeastern Peru, and northern Chile.

Conceptualizes the flow of time in a direction opposite to what is usual for speakers of English and other Indo-European languages.

They speak of **the future being behind them**, while the <u>past is in front of them</u>.





ASIDE: (some forms of the) future tense in English

① **Future Simple**: will (modal) base form of the verb

- Used to describe (or predict) a simple action that will happen in the future:
  - She **will** talk to you later.
  - If you touch that you **will** get hurt.

② **Future**: be going to

- Used to talk about action in the future that are certain and have already been decided.
  - They **are going to** sing tonight.

③ **Future Continuous**: will (modal) be (verb + ing)

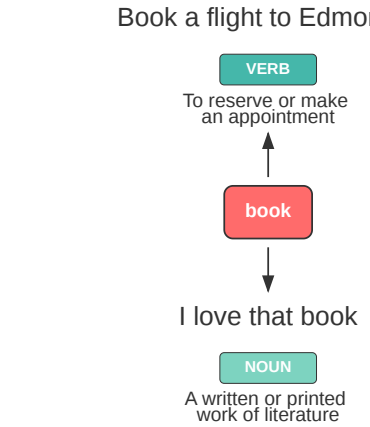
- Used to describe something that we know will be in progress in the future.
  - I **will be taking** classes next term.

PoS and computer "understanding"

Generally, we cannot assign meaning to individual words in isolation. Instead, agents need to "interpret" sentences or phrases as units.

- Look for a combination of **parts of speech** that can be assigned to individual words.
- Look for **grammar rules** that, together, can assign meaning to the whole sentence.

Ambiguity of "book": Verb vs Noun



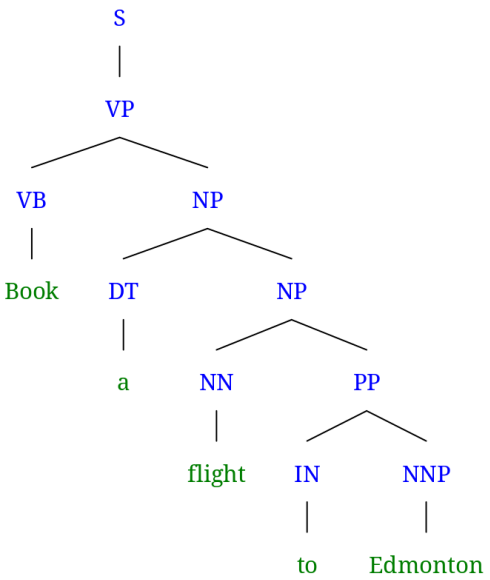
Lexical Ambiguity
The same word form can have different meanings and grammatical functions. Context determines whether "book" functions as a verb (action) or noun (object).

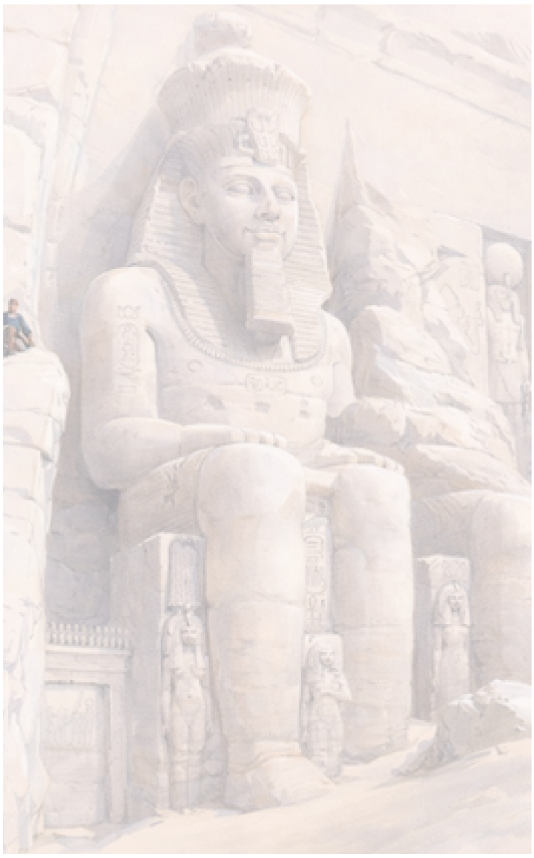
Dealing with ambiguity: parsing

"Classical" NLP methods were based on linguistic knowledge:

- The **imperative** is a grammatical mood that forms a command or request.
- **In English**, the imperative is formed using the infinitive form of the verb.

> The token `book` corresponds to the infinitive form of a verb!





Commonly used (English) Parts of Speech

Commonly used (English) Parts of Speech

	examples	meaning
Noun	dog, love, soccer, fortnight	represents a concrete or abstract thing
Verb	eat, think, love, become	conveys an action, an occurrence, or a state of being
Adjective	good, funny, interesting	word that describes or defines a noun (phrase)
Adverb	greatly, widely, I ate **yesterday**	word or expression that modifies a verb, an adjective, another adverb, a determiner, a clause, a preposition, or a sentence

Commonly used (English) Parts of Speech

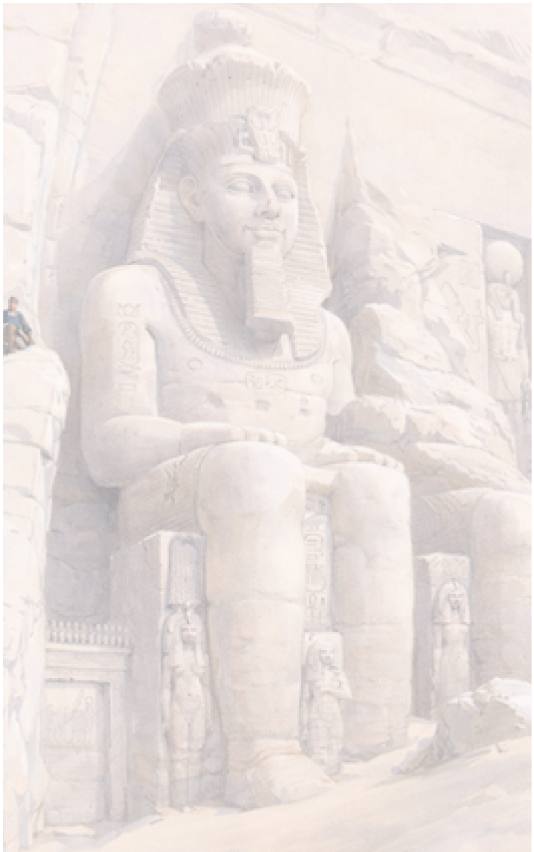
	examples	meaning
Participle	eating, gone, cleaned	a verb form that has some of the characteristics and functions of both verbs and adjectives
Article	a, the, no	words that are used with noun (phrases) to mark the identifiability of the referents of the noun (phrases)
Pronoun	I, it, they	a word or a group of words that one may substitute for a noun (phrase)
Preposition	in, of, until	express spatial or temporal relations (among others)
Conjunction	and, but, although	connects words, phrases, or clauses

PoS tagging by hand

Assigning PoS "tags" to words is an important (preprocessing) task in NLP.

	The	nice	koala	put	the	keys	on	the	table	quickly
N (noun)										
V (verb)										
J (adjective)										
RB (adverb)										
DT (determiner)										
IN (preposition)										

	The	nice	koala	put	the	keys	on	the	table	quickly
N (noun)			✅			✅			✅	
V (verb)				✅						
J (adjective)		✅								
RB (adverb)										✅
DT (determiner)	✅				✅			✅		
IN (preposition)							✅			



Open and closed word classes

Content words and *Function* words

Content words, in linguistics, are words that possess semantic content and contribute to the meaning of the sentence in which they occur.

> In a traditional approach, **nouns** were said to name objects and other entities, **lexical verbs** to indicate actions, **adjectives** to refer to attributes of entities, and **adverbs** to attributes of actions.

They contrast with **function words**, which have very little substantive meaning and *primarily denote grammatical relationships between content words*.

> Examples: as **prepositions** (in, out, under etc.), **pronouns** (I, you, he, who etc.) and **conjunctions** (and, but, till, as etc.).

Slightly edited from https://en.wikipedia.org/wiki/Content_word

Content or function?

Content words can be replaced by other words and still produce a meaningful sentence.

Example: `I $x$ my $y$ over the $z$ yesterday.`

`$x$`	`$y$`	`$z$`
walked	dog	bridge
drove	car	bump
spoke	mind	noise

Closed classes

Relatively small, **fixed set** of *function words* (i.e., short common words which play a role in grammar).

Prepositions: of, in, by, …

Auxiliaries: may, can, will, had, been, …

Pronouns: I, you, she, mine, his, them, …

Modals: will, may, …

Determiners: the, a, an, …

Conjunctions: and, but, yet, …

Generally, words in these classes have a single **PoS**.

Tagging closed-class words

There are a few hundred function words in English and most of them have a clear meaning, in context.

However: Is "up" a particle or an adverb?

- Sammy eats up his dinner.
- Suzan eats her dinner up.

There are other examples...

Particles are function words that must be associated with another word or phrase to impart meaning and that do not satisfy definitions of other universal parts of speech.

Open Classes

Open class: **large number** of *content words* and new ones are easily invented.

- Often, words in one class derive new words for another class. Ex: the noun Google derived the verb "to google"!

English has four:

Nouns: Google, googler

Adjectives: geeky, ridonculous

Verbs: google

Adverbs: automagically

> It is believed that all languages have at least nouns and verbs while most have all four...

Open class words: nouns

Proper nouns

- Are unique (University of Alberta, Ratatouille, Apple, Edmonton Transit Service)
- English capitalizes these
- Many have abbreviations (e.g., ETS)

Common nouns

- Are not unique (ratatouille, apple)
- German capitalizes these.

Count nouns

- Are countable
- Have plurals: one goat, two goat**s**

Mass Nouns:

- Not countable. Ex: fish, salt, communism

> "two fish**es**": two fish of different species or groups.

Open class words: verbs

Convey actions (walk, ate), occurrences (it rained), or states (become, seem).

In English (and other languages) take morphological affixes (eat / eat**s** / eat**en**).

- Can reflect number, tense, ...

Regular verbs follow the same rules for forming the *past* and *past participle*:

- ask/ask**ed**/ask**ed**
- start/start**ed**/start**ed**

Irregular verbs don't!

- do/did/done
- arise/arose/arisen
- have/had/had

Open class words: adjectives and adverbs

Adjectives: identify properties or qualities of nouns (modify nouns)

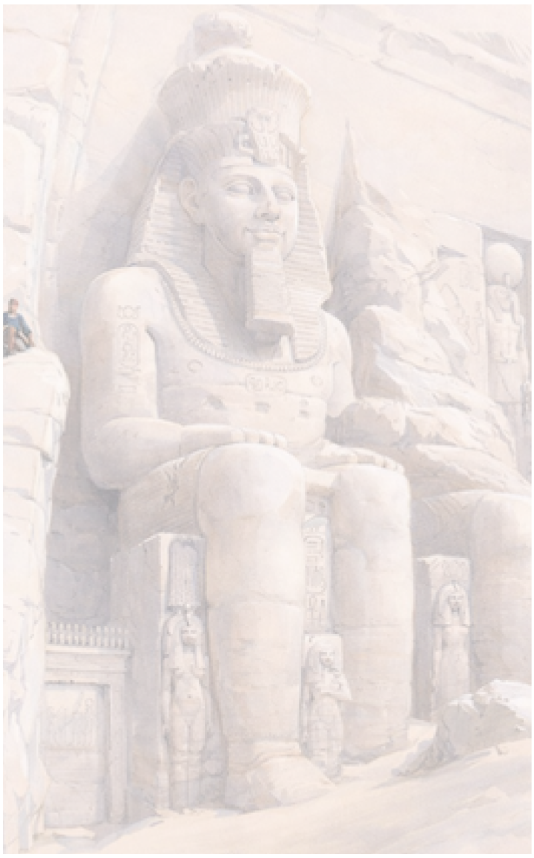
- Color, size, age, …

Adjective ordering restrictions in English:

- Old blue book, not Blue old book

Adverbs: also modify things (verbs, adjectives, adverbs)

- Directional/locative adverbs (here, home, downhill)
- Temporal (Monday, tomorrow)
- Degree adverbs (extremely, very, somewhat)
- Manner adverbs (slowly, slinkily, delicately)



Computer programs to tag words

- English PoS tagsets
- Transducers
- The TBL (Brill) Tagger

Tagsets

Humans (at least in the Western world) relied on eight PoS for over 2,500 years

- Noun, adjective, verb, adverb, participle, article, pronoun, preposition, conjunction.

In the last century, however, linguists have created many more parts of speech, often by subdividing the "core" classes based on word function:

- Ex: Proper nouns, singular nouns and plural nouns (instead of just nouns).

Unfortunately, not all linguists agree on how to refine the "core" PoS, leading to different tagsets:

- Universal Dependencies (https://universaldependencies.org/u/pos/) has 17 tags
- The Penn Treebank has 36

The Penn Treebank

https://www.ling.upenn.edu/courses/Fall_2003/ling001/penn_treebank_pos.html

Tag	Description	Example	Tag	Description	Example	Tag	Description	Example
CC	coord. conj.	*and, but, or*	NNP	proper noun, sing.	*IBM*	TO	infinitive to	*to*
CD	cardinal number	*one, two*	NNPS	proper noun, plu.	*Carolinas*	UH	interjection	*ah, oops*
DT	determiner	*a, the*	NNS	noun, plural	*llamas*	VB	verb base	*eat*
EX	existential 'there'	*there*	PDT	predeterminer	*all, both*	VBD	verb past tense	*ate*
FW	foreign word	*mea culpa*	POS	possessive ending	*'s*	VBG	verb gerund	*eating*
IN	preposition/ subordin-conj	*of, in, by*	PRP	personal pronoun	*I, you, he*	VBN	verb past participle	*eaten*
JJ	adjective	*yellow*	PRP$	possess. pronoun	*your*	VBP	verb non-3sg-pr	*eat*
JJR	comparative adj	*bigger*	RB	adverb	*quickly*	VBZ	verb 3sg pres	*eats*
JJS	superlative adj	*wildest*	RBR	comparative adv	*faster*	WDT	wh-determ.	*which, that*
LS	list item marker	*1, 2, One*	RBS	superlatv. adv	*fastest*	WP	wh-pronoun	*what, who*
MD	modal	*can, should*	RP	particle	*up, off*	WP$	wh-possess.	*whose*
NN	sing or mass noun	*llama*	SYM	symbol	*+, %, &*	WRB	wh-adverb	*how, where*

Figure 17.2 Penn Treebank core 36 part-of-speech tags.

Computer programs for tagging words

Different kinds of computer systems have been created for the task of PoS tagging.

Classical systems relied on transducers that would recognize words in a given class.

- One transducer for each class! (ex: one for verbs, another for nouns).
- Linguists would provide the lists of words for each class! Computer Scientists would implement the systems.

In the 1980s **machine learning** systems were developed to learn how to tag words from examples

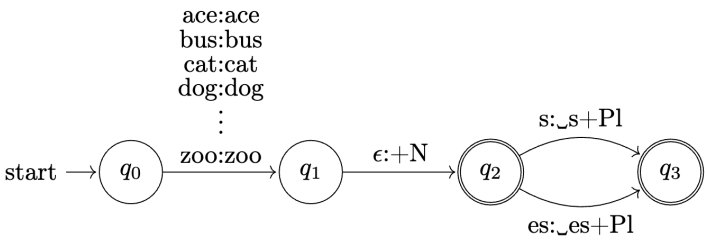
- The TBL (Brill) tagger is a famous example.

Today, LLMs do not seem to perform PoS tagging at all.

Transducers for PoS tagging

Transducers have been used in early systems for recognizing words that are members of strictly regular word classes.

- For example, the following transducer recognizes quite a few nouns in their base form or in their plural form, when the plural is formed by appending `s` or `es`:



Input: `cars`; output `car+N` `s+Pl`

Transducers for PoS tagging

Pros:
FSA are fast and "easy" to program (there are many libraries for that).

We can easily debug FSA.

FSA are **composable**. We can concatenate FSA that recognize the root of a word with others that recognize known suffixes:

- base noun + `s` for plural
- base verb + `ed` for simple past

Cons:
Transducers require the list of words in each class (difficult to get for open classes).

Transducers work well with **regular** languages. Many word classes have many exceptions:
- talk/talked/talked; eat/ate/eaten
- dog/dogs; goose/geese; moose/moose

What to do if multiple transducers recognize the same word?

NLTK regex-based Part of Speech tagger

The `RegexpTagger` associates PoS tags to regular expressions. The word is assigned to the tag corresponding to the **first** regular expression that matches the word.

- Ex: `buses` is tagged as `VBZ` because it ends in **es**

```
patterns= [(r'.*ing$', 'VBG'),       # gerunds
  (r'.*ed$', 'VBD'),                 # simple past
  (r'.*e?s$', 'VBZ'),                # 3rd singular present
  (r'.*ould$', 'MD'),                # modals
  (r'.*s$', 'NNS'),                  # plural nouns
  (r'^-?[0-9]+(\.[0-9]+)?$', 'CD'),  # cardinal numbers
  # more patterns go here
  (r'.*', 'NN')                      # nouns (default)
  ])
regexp_tagger = RegexpTagger(patterns)
regexp_tagger.tag("I won the race")
```

https://www.nltk.org/api/nltk.tag.sequential.html#nltk.tag.sequential.RegexpTagger

Ambiguity in PoS tagging

Studies on news corpora (WSJ-Wall Street Journal and a corpus from Brown University) show that about half of the words have a unique PoS!

When we **normalize** words into a common lemma (ex: working → work; worked → work) the ambiguity drops considerably.

This means that we need only to find the correct tag for about half of the words.

	Unambiguous (1 Tag)		Ambiguous (2+ tags)	
	Count	%	Count	%
Types				
WSJ	44 432	86	7 025	14
Brown	45 799	85	8 050	15
Tokens				
WSJ	577 421	45	711 646	55
Brown	384 349	33	786 646	67

"Type" means normalized word (ex: via lemmatization).

"Token" means, essentially, individual word.

Some common words are (very) ambiguous

Example: Penn Treebank tags for the word **back** :

Tag	Description	Example
JJ	Adjective	earnings growth took a **back** seat
NN	Singular or mass noun	a small pain in the **back**
VBP	Verb, non-3rd person singular present	a clear majority of senators **back** the bill
VB	Verb, base form	Dave began to **back** toward the door
RP	Particle	enable the country to buy **back** debt
RB	Adverb	I was twenty-one **back** then

Statistical tagging

The **Most Frequent Class Baseline** works by assigning each token to the class it occurred in most often in the training set.

Example:

Looking at occurrences of **race** in the Brown corpus, the Baseline would be:
P(**NN**|race) = .98 P(**VB**|race)= .02

Resulting in the following labeling:

✅ I **/PRP** won **/VBD** the **/DT** race **/NN** yesterday **/RB**
❌ I **/PRP** expect **/VB** to **/TO** race **/NN** tomorrow **/NN**

The TBL (Brill) Tagger

The "Transformation Based Learning" tagger (Brill, 1994) was one of the first successful machine learning PoS taggers.

It first tags words (in a training corpus) using the Most Frequent Class.

Next, the learner looks for ways of fixing incorrect tags according to the ground truth annotations.

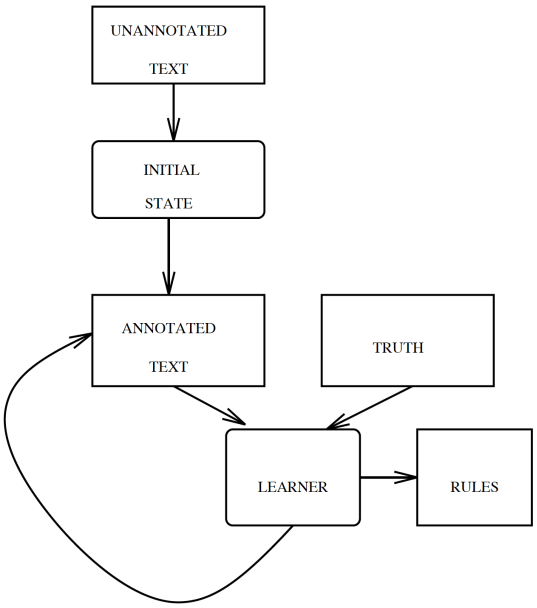


Figure 1: Transformation-Based Error-Driven Learning.

Learning transformation rules

The learner applies the Most Frequent Class first, and then proceeds iteratively:
At iteration i it looks for transformation rules like this (more are possible):

Transform tag **X** to **Y** if the previous/following word is **W**

Example:

Training sentence: I/PRP expected/VBN to/TO race **/VB** tomorrow/NN

Tags at iteration i: I/PRP expected/VBN to/TO race **/NN** tomorrow/NN

Learned rule: "Transform **NN** to **VB** if the previous word is `to`"

Which transformations do we keep?

At each iteration, keep the transformation that minimizes the number of tagging errors in the training data.

- Break ties arbitrarily?
- Backtrack in case of ties to find the optimal set of rules?

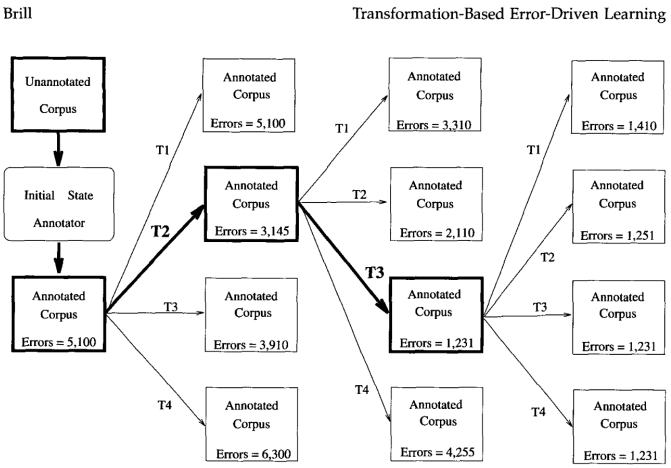


Figure 2
An Example of Transformation-Based Error-Driven Learning.

Dealing with unknown/unseen words

What can a tagger do when it encounters a word never seen before (in the training data or in the lists of words created by the linguists)?

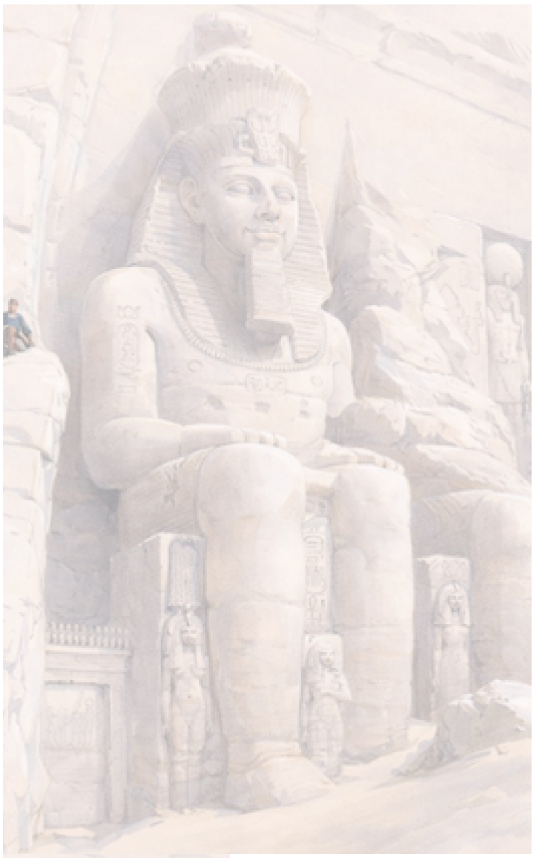
- Ex: Tupi **/UNKN** is a language from the Tupi-Guarani **/UNKN** family.

A transducer-based tagger would fail with unknown words!

NLTK's `RegexpTagger` tagger could assign unknown words to `NN` (noun) if the word did not match any rule. But it could also assign an incorrect tag.

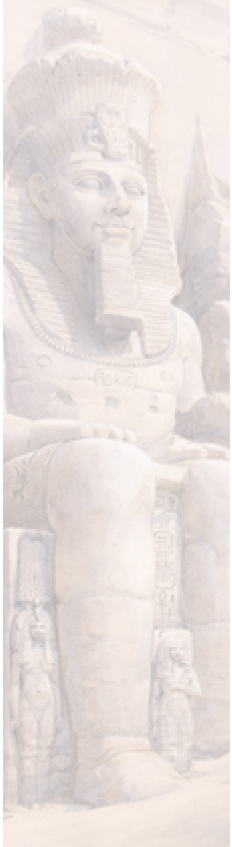
The **TBL tagger can learn to tag unknown words correctly**!

- Example rule "Transform UNKN to NN if the following word is `is`"



Relationship to other modules

1. Regular expressions
2. Finite-state Transducers
3. Parsing, ambiguity
4. Hidden Markov Models (HMMs) for PoS Tagging/sequence labelling
5. Named Entity Recognition



What you should know or be able to do

- Tag English texts using standard tagsets such as the Penn Treebank
- Identify ambiguous words and explain the source of such ambiguity
- Explain how transducers can be created for PoS tagging
- Explain limitations of lexicon-based taggers that use transducers
- Explain limitations of tagging by the most frequent tag in the corpus
- Explain the regular-expression based tagger
- Explain how the TBL tagging algorithm works and how it learns
- Explain how a given transformation rule in the TBL algorithm could have been learned